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Relaxing Intertemporal Separability: A Rational Habits Model of Labor Supply Estimated from Panel Data

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This article relaxes the assumption of separability of preferences over time that, although implausible, is usually maintained in life-cycle labor-supply models. An empirically tractable rational habits model is specified and estimated on the basis of a 10-year sample of men from the Michigan Panel of Income Dynamics. The effect of past hours of work in determining current hours decisions is found to be very important and well determined, having allowed for permanent individual differences and for time effects.

I. Introduction

Most of the micro studies on intertemporal labor supply rely on the use of some conditioning variable that summarizes the influence of the relevant variables from outside the current period. For example, in Heckman and MaCurdy (1980), MaCurdy (1981), and Browning, Deaton, and Irish (1985), this variable is the Lagrange multiplier or marginal utility of wealth; in Blundell and Walker (1986), it is the current period's expenditure, and in Altonji (1986), it is current consumption. The wage effect estimated

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holding this summary variable constant is the much-studied intertemporal substitution elasticity, which measures how the worker's labor supply reacts to a change in the wage along a *given* lifetime wage profile. With this approach, relatively simple intertemporal models are obtained even under uncertainty, and there is no need to make any assumptions about future variables, in particular about future wages and interest rates.

However, these models do assume that preferences are separable over time, and therefore hours supplied in period t are independent of hours supplied in period $t - 1$.¹ But one can think of several reasons that make implausible a full adjustment within the period. First, there are search costs that make people act as they did before; second, there are borrowing limitations that in a life-cycle context would impede a worker from working less in a period in which his real wage is low compared with his expected future wage; third, there are contract ties and institutionally fixed hours. Finally, another general reason for nonseparability over time of intertemporal models that has received a good deal of attention in the consumption literature is that tastes are affected by previous experiences of consumption and leisure, that is, "habit formation" (e.g., Pollack and Wales 1969, Phelps and Spinnewyn 1982, and Muellbauer 1986). The first three reasons would lead to nonseparabilities in the budget constraint, while the latter gives rise to nonseparabilities in preferences of the kind that are formally dealt with in this article. All those considerations point in the direction of some stickiness in the labor-supply response and, thus, of the possible relevance of lagged hours in determining current hours.

There has been some recent work in trying to test and relax intertemporal separability in life-cycle labor-supply models using micro data. Hotz, Kydland, and Sedlacek (1988) start from a general utility function under uncertainty that does not impose intertemporal separability; by including habit effects. They examine the importance of nonseparable preferences for the hours decision using the consumption Euler equation. Their results provide quite general and complete information on the nature and the extent of state dependence in the hours of work decision. However, although their findings using a sample of men from the Michigan Panel of Income Dynamics strongly reject separability, they do not estimate a life-cycle labor-supply model allowing for this more general structure of preferences, and no elasticity estimates under nonseparability are available. Johnson and Pencavel (1984) estimate a labor-supply function derived from a model for just two periods with habits, and hence their results are not comparable with the above mentioned literature. However, their estimates of the extent of state dependence are likely to be biased because

¹ Note that Browning (1985) generalizes the model in Browning, Deaton, and Irish (1985) to allow for intertemporal nonseparability under certainty by including past and future period wages in the Frisch labor-supply function.

they do not take into account the correlation between lagged hours and the error term.

The purpose of this work is to test for intertemporal separability and to estimate a rational habits life-cycle labor-supply model for men using the Michigan panel data. The basic framework for my model is a structural specification of intertemporal labor supply where, instead of conditioning on the marginal utility of wealth, λ say, I adopt a specification of preferences that allows me to work with an explicit form for the solved λ , in terms of all future wages, initial assets, interest rate, and utility function parameters. Therefore, in my approach for relaxing intertemporal separability, I require some approximation for future wages and no uncertainty (although I allow for some replanning). An advantage of this method is that I not only estimate the effect of wage changes along a given wage profile, but I am also able to estimate the effect of having different wage profiles (cross-uncompensated elasticity), which is needed for making the usual comparative static exercises. MaCurdy (1981) estimates uncompensated elasticities in a second step, without taking into account the restrictions implied by the optimization problem.

II. A Rational Habits Model

The Model

In this section, I develop an empirical rational-habits model for dynamic intertemporal labor supply.

If tastes are affected by previous leisure-work experience or, in other words, if the individual feels it is costly to vary his labor supply, the habit-persistence literature (e.g., Muellbauer [1986] in an intertemporal context) suggests the following form for the life-cycle individual utility:

$$U_{it} = \sum_{s=0}^{T-t} (1 + \rho)^{-s} U(h_{it+s} - \phi h_{it+s-1}, c_{it+s}), \quad t = 1, \dots, T, \quad (1)$$

where h_{it} is actual hours at t for individual i , c_{it} is the consumption level, ρ is the rate of time preferences, and ϕ measures habit persistence. To allow for interdependence over more than two periods, we could replace ϕh_{it-1} by $\sum_{j=1}^J \phi_j h_{it-j}$. Equation (1) can be regarded as a special case of the more general formulation given by

$$U_{it} = \sum_{s=0}^{T-t} (1 + \rho)^{-s} U[f(h_{it+s}, h_{it+s} - h_{it+s-1}), c_{it+s}], \quad (2)$$

where f is some function of present hours and of the difference between present and past hours worked, which reflects the cost derived from the

adjustment. It is assumed that the person does not perceive that the utility function will be shifted by the change from previous consumption.

Adopting a Stone-Geary specification for the utility function, the individual maximizes

$$\sum_{k=t}^T \frac{1}{(1+\rho)^{k-t}} [B_{1ik} \ln(\gamma_b + \phi h_{ik-1} - h_{ik}) + B_{2ik} \ln(c_{ik} - \gamma_c)], \quad (3)$$

where B_{1ik} and B_{2ik} are age-specific modifiers of tastes, γ_c is the minimum necessary commodity consumption level, and γ_b can be thought as the maximum feasible amount of $(h_{it} - \phi h_{it-1})$. We assume that $B_{1it} + B_{2it} = 1$, which imposes homogeneity. Furthermore, the choice of a Stone-Geary functional form to describe both the utility within one period and the between-periods relation assumes a unitary elasticity of substitution not only between effective consumption and effective hours of leisure (“effective” being defined as the actual minus the committed level) but also between effective hours of leisure in different periods (and similarly for consumption). The budget constraint is

$$\sum_{k=t}^T (1+r)^{t-k} (w_{ik} h_{ik} - c_{ik}) = -A_{it}, \quad (4)$$

where r is the real rate of interest, which is assumed constant, w_{it} is the real wage, and A_{it} is the asset level. The model to be developed is obtained under the assumption of perfect certainty but allows for replanning. That is, the individual at each period t will use the latest information to revise his expected wage path, but no insurance will be taken. Notice that following the present approach there is no obvious extension to allow for general uncertainty.

Defining $h_{ik}^+ = h_{ik} - \phi h_{ik-1}$, we can write the budget constraint as²

$$\begin{aligned} \sum_{k=t}^T (1+r)^{t-k} c_{ik} - A_{it} &= \sum_{k=t}^T (1+r)^{t-k} (w_{ik} h_{ik}^+ + \phi w_{ik} h_{ik-1}) \\ &= (w_{it} h_{it}^+ + \phi w_{it} h_{it-1}) + \frac{1}{1+r} \\ &\quad \times (w_{it+1} h_{it+1}^+ + \phi w_{it+1} h_{it}^+ + \phi^2 w_{it+1} h_{it-1}) + \dots; \end{aligned}$$

combining terms in h_{it}^+ , h_{it+1}^+ , and so forth, we find

$$\sum_{k=t}^T (1+r)^{t-k} (h_{ik}^+ w_{ik}^+ - c_{ik}) = -A_{it} - \phi h_{it-1} w_{it}^+, \quad (5)$$

² See Muellbauer (1985) for similar transformations with consumption.

where

$$w_{ik}^+ = \sum_{j=0}^{T-k} [\phi/(1+r)]^j w_{ik+j}.$$

Maximizing (3) with respect to this transformed budget constraint (5), we have

$$h_{it}^+ = \gamma_h - \frac{B_{1it}}{w_{it}^+} \lambda_{it}^{-1} \quad (6)$$

and

$$c_{it} = \gamma_c + B_{2it} \lambda_{it}^{-1}, \quad (7)$$

where λ_{it} is the Lagrange multiplier. Substituting these into the budget constraint and using the assumption that $B_{1ik} + B_{2ik} = 1$, we obtain

$$\lambda_{it}^{-1} = \left[\sum_{k=t}^T (1+\rho)^{t-k} \right]^{-1} \left[\sum_{k=t}^T (1+r)^{t-k} (\gamma_h w_{ik}^+ - \gamma_c) + A_{it} + \phi h_{it-1} w_{it}^+ \right]. \quad (8)$$

In a certainty model, the w_{it} for all future periods are assumed to be known at each t . However, this poses an empirical problem since not all wages are actually observed.

Future wages are assumed to be on an individual specific time trend. I allow for wage replanning by assuming that each period agents form new predictions about their future wage profiles. Here I assume a linear trend, thus having

$$\hat{w}_{ik}^t = \hat{\alpha}_{it} + \hat{\delta}_{it} k, \quad k = t, \dots, T, \quad (9)$$

where, at each period t , individuals make use of their most recent observations to determine $\hat{\alpha}_{it}$ and $\hat{\delta}_{it}$.³

³ Specifically, the last n observations available at each period t are used to calculate

$$\hat{\delta}_{it} = \frac{1}{n-1} \sum_{k=2}^n \Delta w_{i(t-k+2)} = \frac{1}{n-1} (w_{it} - w_{it-n+1})$$

and then

$$\hat{\alpha}_{it} = \frac{1}{n} \sum_{k=1}^n (w_{it-k+1} - \hat{\delta}_{it} k) = w_{it}^t - \frac{n+1}{2} \hat{\delta}_{it},$$

Further, assuming an infinite horizon and that $\rho = r$, we obtain⁴

$$\lambda_{it}^{-1} = \left(\frac{1+r}{1+r-\varphi} \right) \gamma_b \left[\hat{\alpha}_{it} + \hat{\delta}_{it} \left(t + \frac{1}{r} + \frac{\varphi}{1+r-\varphi} \right) \right] - \gamma_c + \frac{r}{1+r} A_{it} \\ + \left(\frac{r}{1+r-\varphi} \right) \varphi h_{it-1} \left[\hat{\alpha}_{it} + \hat{\delta}_{it} \left(t + \frac{\varphi}{1+r-\varphi} \right) \right]. \quad (10)$$

Multiplying both sides of (6) by w_{it} and again replacing future values of w_i by their approximations at t , the denominator in (6) is equal to

$$\frac{1+r}{1+r-\varphi} \frac{\left[\hat{w}_{it} + \left(\frac{\varphi}{1+r-\varphi} \hat{\delta}_{it} \right) \right]}{w_{it}} \simeq \frac{1+r}{1+r-\varphi}, \quad (11)$$

given the typically negligible value of $(\varphi/1+r-\varphi)\hat{\delta}_{it}$ relative to $\hat{w}_{it}$ (where $\hat{w}_{it}$ is $\hat{w}_{it}^t$ as defined in [9]).

We assume that B_{iit} varies with observable and unobservable personal characteristics, and this can be represented in the form

$$B_{iit} = \beta z_{it} + \zeta_{it}, \quad (12)$$

where z_{it} is a column vector of observed exogenous variables, β is a row vector of parameters to be estimated, and ζ_{it} is a random error. The unobserved random component of B_i as well as the unobserved variability of γ_b will be subsumed in the equation error term. I believe that, in the case of men, observable individual differences are more likely to affect B_{iit} , which is the weight given to leisure, rather than γ_b , which is the maximum quasi difference in number of hours. Therefore the final expression for the rational-habits earnings equation is⁵

where w_{it}^t is the time mean of the most recent n observations on w for the i th individual at period t . (See MaCurdy [1981] for a somewhat similar approach.) The choice of n depends on the number of time-series observations in the sample. Here I use $n = 4$.

⁴ Notice that the assumption $\rho = r$ is innocuous in my model since a separate effect of ρ is not identified. Without this assumption, the expression for λ_{it}^{-1} in (10) is multiplied by the constant $\rho(1+r)/[(1+\rho)r]$.

⁵ This rational model is empirically undistinguishable from certain myopic models. In particular, in a previous version of this article (Bover 1986) I considered a myopic-habits model in which the intertemporal utility was specified to be

$$U_{it} = \sum_{s=0}^{T-t} (1+\rho)^{-s} U(h_{it+s} - \varphi h_{it-1}, c_{it+s}),$$

$$h_{it}\bar{w}_{it} = \gamma_b \bar{w}_{it} + \phi h_{it-1} \bar{w}_{it} + \mu_1 z_{it} + \mu_2 z_{it} \hat{w}_{it} + \mu_3 z_{it} \hat{\delta}_{it} + \mu_4 z_{it} A_{it} + \mu_5 h_{it-1} z_{it} \hat{w}_{it} + \mu_6 h_{it-1} z_{it} \hat{\delta}_{it}, \quad (13)$$

where

$$\begin{aligned} \mu_1 &= \beta \gamma_c (1 + r - \phi) / (1 + r), \\ \mu_2 &= -\beta \gamma_b, \\ \mu_3 &= -\beta \gamma_b [r^{-1} + \phi / (1 + r - \phi)], \\ \mu_4 &= -\beta r (1 + r - \phi) / (1 + r)^2, \\ \mu_5 &= -\beta \phi r / (1 + r), \end{aligned}$$

and

$$\mu_6 = \beta \phi^2 r / [(1 + r)(1 + r - \phi)].$$

In this model, the effect of lagged hours on current hours is measured by

$$\phi \left[\bar{w}_{it} - \frac{r}{1+r} \beta(z_{it} \hat{w}_{it}) - \frac{r}{1+r} \frac{\phi}{1+r-\phi} \beta(z_{it} \hat{\delta}_{it}) \right] / \bar{w}_{it}. \quad (14)$$

Elasticities

The elasticity that has been most studied is the one that measures the effect of current wage variations holding λ_{it} , the marginal utility of wealth, constant. That is, controlling for the individual's wage profile and assets level, how much more he will work during the periods when his wage is relatively higher? This elasticity, often called the intertemporal labor-supply elasticity, in this model is given by

and a partial adjustment model where

$$(h_{it} - h_{it-1}) = \psi(h_{it}^* - h_{it-1}),$$

h_{it}^* being the desired level of hours of work obtained from a one-period optimization model. The three models lead to similar estimating equations.

$$\begin{aligned}
 \epsilon_{it} &= \frac{w_{it}}{h_{it}} \frac{\partial h_{it}}{\partial w_{it}} \Big|_{\lambda_{it} = \text{constant}} \\
 &= B_{lit} \frac{1+r-\varphi}{1+r} \lambda_{it}^{-1} / h_{it} w_{it} \\
 &= \left(\frac{\gamma_b + \varphi h_{it-1}}{h_{it}} \right) - 1.
 \end{aligned} \tag{15}$$

However, ϵ_{it} alone does not suffice for the usual comparative static exercises to be made. Indeed, for this we need to know what effects wage variations will have on hours *through* their impact on λ . That is, what is the effect on labor supply of having *different* wage profiles? Given the assumption of a linear trend for each individual's wage profile, we may distinguish intercept and slope differences. Having wage profiles that only differ by a constant will lead to different marginal utilities of wealth (λ_{it}), and this will result in hours variations in all periods. This effect is measured by the following elasticity:

$$\begin{aligned}
 \eta_{it}^a &= \frac{\alpha_{it}}{h_{it}} \frac{\partial h_{it}}{\partial \lambda_{it}} \frac{\partial \lambda_{it}}{\partial \alpha_{it}} \\
 &= -B_{lit} \left(\gamma_b + \varphi \frac{r}{1+r} h_{it-1} \right) \frac{\alpha_{it}}{h_{it} w_{it}} \\
 &= \left[1 - \left(\frac{\gamma_b + \varphi h_{it-1}}{h_{it}} \right) \right] \left(\gamma_b + \varphi \frac{r}{1+r} h_{it-1} \right) \lambda_{it} \frac{1+r}{1+r+\varphi} \alpha_{it}.
 \end{aligned} \tag{16}$$

Moreover, for a different slope the effect on hours through the influence of λ is given by

$$\begin{aligned}
 \eta_{it}^s &= \frac{\delta_{it}}{h_{it}} \frac{\partial h_{it}}{\partial \lambda_{it}} \frac{\partial \lambda_{it}}{\partial \delta_{it}} \\
 &= -B_{lit} \left[t \left(\gamma_b + \varphi \frac{r}{1+r} h_{it-1} \right) + \gamma_b \left(\frac{1}{r} + \frac{\varphi}{1+r+\varphi} \right) \right. \\
 &\quad \left. + \frac{\varphi^2 r}{(1+r)(1+r-\varphi)} h_{it-1} \right] \frac{\delta_{it}}{h_{it} w_{it}} \\
 &= \left[1 - \left(\frac{\gamma_b + \varphi h_{it-1}}{h_{it}} \right) \right] \left[t \left(\gamma_b + \varphi \frac{r}{1+r} h_{it-1} \right) + \gamma_b \left(\frac{1}{r} + \frac{\varphi}{1+r-\varphi} \right) \right. \\
 &\quad \left. + \varphi \frac{r}{1+r} \frac{\varphi}{1+r-\varphi} h_{it-1} \right] \lambda_{it} \frac{1+r}{1+r-\varphi} \delta_{it}.
 \end{aligned} \tag{17}$$

The elasticities given in (16) and (17) are called in the literature “cross-uncompensated elasticities.” So far, I have stated the elasticity to a wage variation holding λ constant and the elasticities to wage-profile variations through the influence of λ on hours. To obtain the *total* effect on h_{it} of wage variations, we can use combinations of ε_{it} and η_{it}^a, η_{it}^b .⁶ Notice that, since we are in a perfect foresight context, that is, lifetime paths of wages are assumed to be known for each individual, we can only consider differences in the wage profile between different individuals or shifts in the wage profile of an individual due to replanning.

All the elasticities described above are short-run elasticities in the sense that they do not include the lasting impact of wage changes through the effect of h_{it-1} . To obtain long-run elasticities, we should take into account the effect of h_{it-1} on h_{it} .

Note that for each elasticity ([15], [16], and [17]) two expressions are reported. The first ones are obtained from straight application of the elasticities definitions. The second ones, which are usually reported in the literature dealing with Stone-Geary specifications, are obtained using the equality $h_{it}^+ = \gamma_b - B_{1it}(1 + r - \phi/1 + r)\lambda_{it}^{-1}$. However, if we allow for individual and/or time effects on the right-hand side of the above equality, the two expressions cease to be equivalent. The elasticities estimated in the second way will be somewhat biased, the size of the bias depending on the importance of the individual and time effects. Therefore the elasticities I report are evaluated using the first expressions.

III. Estimation

Equation (13) is an earnings equation for individual i at time t of the form

$$y_{it} = x'_{it}(x_{it}^*)\pi(\theta) + u_{it}, \quad i = 1, \dots, N; t = 1, \dots, T, \quad (18)$$

where y_{it} is real earnings of the i th individual in the t th year, x_{it} is a $k \times 1$ vector of regressors that are functions of some original variables x_{it}^* (e.g., x_{it} includes terms $(h_{it-1}w_{it}), (z_{it}\hat{w}_{it}), \dots$, whereas in x_{it}^* we have $h_{it-1}, z_{it}, \dots$), $\pi(\theta)$ is a $k \times 1$ vector of coefficients that are functions of the parameters of interest θ , for example, $\theta = (\gamma_b \gamma_c \phi r \beta)$. We have therefore an equation that is nonlinear in the parameters but linear in some transformation of the original variables.

The error term u_{it} can be interpreted as unobserved heterogeneity related to B_1 and γ_b and also as a forecast error due to the approximation used for future real wages. The error scheme I adopt is

$$u_{it} = f_i + v_{it}, \quad (19)$$

⁶ For more details on these elasticities see MaCurdy (1981).

where f_i is a time-invariant individual effect. The component f_i captures the effects of unobserved permanent differences in tastes and background across individuals that are likely to be correlated with the observed individual's characteristics. I will consider the variables in their deviations from time means (i.e., a within-groups type of solution) in order to control for the correlation between the regressors and the individual effects. Note that to account for individual effects is specially important since the presence of unobserved heterogeneity in dynamic models may bias one's findings concerning the presence of state dependence as it induces spurious correlation between current and past values of the endogenous variables. The v_{it} are assumed to be independently distributed across individuals but not necessarily over time. For example, the presence of unobserved variables that vary systematically over the sample period could cause the v_{it} to be serially correlated.

In a vector notation, stacking the earnings equations given by (18) and assuming from now on that all the variables are in their time mean deviations, we have

$$y = X\pi(\theta) + v, \quad (20)$$

where

$$y' = (y_{11} \dots y_{1T} \dots y_{N1} \dots y_{NT}),$$

$$X' = (x_{11} \dots x_{1T} \dots x_{N1} \dots x_{NT}),$$

$$v' = (v_{11} \dots v_{1T} \dots v_{N1} \dots v_{NT}),$$

with

$$E(vv') = (I_N \otimes \Omega),$$

where Ω is a $T \times T$ matrix and I_N is the $N \times N$ identity matrix. Since the v 's are in deviations from time means, the matrix Ω is singular (rank $T - 1$). Hence, there is no loss of information by using only $(T - 1)$ of the T observations in time mean deviations, so that the last observation for each individual will be dropped. Notice that in this context there is no need to assume any specific form for the autocorrelation structure of the error term, but rather we can work with an unrestricted intertemporal covariance matrix Ω . Since we have N observations for each t , and T is small relative to N , the ω_{ts} element of Ω can be estimated as

$$\omega_{ts} = \frac{1}{N} \sum_{i=1}^N \hat{v}_{it} \hat{v}_{is}, \quad t, s = 1, \dots, T - 1, \quad (21)$$

where the $\hat{v}_{ik}$'s are the residuals calculated from some initial consistent estimates. The wage variable cannot be treated as exogenous both because it has potential sources of correlation with the error term (see Nickell [1986] for a detailed discussion) and also because it has been repeatedly shown to be subject to an important measurement error (e.g., Abowd and Card 1983; Altonji 1986). But apart from wages, the lagged hours of work variable is also correlated with the error term. Clearly, despite the within-groups transformation, this variable is still correlated with the transformed error term through the mean component even if v_{it} is not serially correlated (see, e.g., Nickell 1981).

Notice that, if we were able to rule out the possibility of serial dependence in the v_{it} on a priori grounds, it would be advantageous to remove the permanent effects f_i by taking first differences instead of deviations from time means because, in such a case, hours of work dated $(t - 2)$ or less could be used as additional instruments in the equation in first differences for period t (e.g., Holtz-Eakin, Newey, and Rosen 1988). However, since we have no good reason to restrict the autocovariances of the errors, it is important to try to identify ϕ and the other structural parameters by the mean of instruments that are exogenous rather than predetermined. Otherwise the estimated effect of lagged hours on current hours could just be a reflection of serial correlation as opposed to genuine habits. These estimates are obtained by minimizing the following generalized instrumental variable criterion function:⁷

$$C(\theta) = v'(I_N \otimes \Omega^{-1})Z[Z'(I_N \otimes \Omega^{-1})Z]^{-1}Z'(I_N \otimes \Omega^{-1})v, \quad (22)$$

where Z is a matrix of exogenous variables which has at least rank p , where p is the total number of parameters to be estimated. In the absence of misspecification, the resulting estimates will be asymptotically normally distributed, and their asymptotic covariance matrix can be estimated by

$$\hat{\Sigma} = \left\{ \frac{\partial \hat{\pi}'}{\partial \theta} X'(I_N \otimes \Omega^{-1})Z[Z'(I_N \otimes \Omega^{-1})Z]^{-1}Z'(I_N \otimes \Omega^{-1})X \frac{\partial \hat{\pi}}{\partial \theta'} \right\}^{-1}.$$

Moreover, the minimized criterion $\hat{C} = C(\hat{\theta})$ will be asymptotically distributed as a chi-square with $(q - p)$ degrees of freedom, where q is the rank of Z . The statistic $\hat{C}$ is the test of overidentifying restrictions (see Sargan 1958) and is a test of the validity of the instruments in the following sense: if $q - p = 0$, θ is just identified and $\xi = Z'(I_N \otimes \Omega^{-1})\hat{v} = 0$, whereas, if $q - p > 0$, there are $q - p$ linear combinations of ξ that are

⁷ A FORTRAN program to minimize (22) was written. I used a modified Newton algorithm as implemented in NAG subroutine E04KDF (NAG Library of Computer Programs, Mark IX, Oxford University Computing Library, Oxford). First derivatives are required, but they were approximated by numerical first differences.

different from zero in the sample but should be equal to zero in the population; these are precisely the restrictions that $\hat{C}$ is testing.

IV. Empirical Analysis

To complete the specification of the model, I have to select a set of instruments Z and a set of variables B_1 that determine tastes for leisure. I also have to define a proxy for the consumer's wealth.

I include as instruments all the exogenous variables that appear in (13) and other variables that might sensibly help in predicting w_{it} and h_{it-1} such as age squared, interaction between education and age, and dummy variables for each year of the sample. A demand-side instrument, the state-unemployment rate, was always included.

The list of instruments includes predicted wages $\hat{w}_{it}$ and the slopes $\hat{\delta}_{it}$ corresponding to the individual wage paths, as well as cross products of these variables with individual characteristics (see the notes to table 1). The fitted-wage paths are smoothed versions of the observed series in which measurement errors would tend to cancel out. However, since the time dimension of the data set is small, so is the feasible length of the moving averages to construct the series of intercepts and slopes. This could cause nonnegligible correlations between the instruments and the errors and thus bias my estimates. If this is the case, it is hoped that the test statistics of overidentifying restrictions would indicate the extent of the problem.

As for the choice of personal characteristics in B_1 , the final specifications included the number of children and a constant.

The data set does not contain any measure of asset level. I use as a proxy the total family income minus total labor income of the family.

The data come from the random sample of the University of Michigan Panel Study of Income Dynamics spanning from 1968 to 1977. The state unemployment variable is the "Annual Average Unemployment Rates, as a percent of Labor Force by State" series (source: Employment & Training Report of the President, 1976–80). Since, for the period previous to 1970, the unemployment variable was calculated using a different method, the sample period spans from 1970 to 1976. But this involves using 9 years of the Michigan panel from 1969 to 1977 since some variables, such as hours of work and labor income, refer to the previous period and because of the presence of lagged hours. Moreover, the first two waves containing observations on hours and earnings for 1967 to 1968 are used in constructing individual wage paths. My sample consists of 785 white men, who were 20–50 years old in 1968, who were not retired, disabled, or full-time students, and who reported positive annual hours and wages for all periods. Individuals with a number of hours per year greater than 5,000 in any period were deleted. Descriptive statistics of the relevant variables for this sample are presented in table A1 in the Appendix.

Table 1 shows the estimates for the utility function parameters, the rate

Table 1
Instrumental Variables Estimates of the Rational-Habits Model
of Labor Supply

	(1)	(2)
γ_b	1,884.8818 (419.49)	1,768.1516 (384.04)
γ_c	6,780.6880 (3,645.55)	4,454.0084 (5,779.59)
ϕ	.3478 (.10)	.2205 (.09)
ρ, r	.4666 (.26)	.2429 (.28)
β_{const}	.3284 (.11)	.1845 (.17)
β_{children}	-.0324 (.005)	-.0238 (.01)
TD1	...	-57.4968 (107.12)
TD2	...	-36.8210 (127.05)
TD3	...	208.3264 (94.65)
TD4	...	274.6359 (105.11)
TD5	...	-7.6614 (111.6614)
TD6	...	-197.3619 (89.43)
ε	.1574	.0822
η^a	-.2576	-.1093
η^b	-.0330	-.0184
Total effect of h_{it-1} on h_{it}	.3195	.2143
Total effect of no. of children on h_{it}	30.4611	31.0055
Chi-squared test of overidentifying restrictions (df in parentheses)	39.93 (12)	11.56 (6)
% of unexplained variation	10.47	9.34

NOTE.—Instruments used: $\hat{w}_{it}$, $\hat{\delta}_{it}$, ($\text{age} \times \hat{w}_{it}$), ($\text{age} \times \hat{\delta}_{it}$), ($\text{age} \times A_{it-1}$), number of children, ($\text{no. of children} \times \hat{\delta}_{it}$), ($\text{no. of children} \times A_{it-1}$), age squared, ($\text{age} \times \text{education}$), ($\text{age} \times \text{education squared}$), state unemployment, six time dummies. Standard errors are in parentheses.

of time preference, and the habit coefficient. It also shows the estimates for the different elasticities, the total effect on hours of the number of children, and the total effect of the lagged hours on current hours; these are data dependent, and the numbers shown in table 1 are evaluated at sample means. In the second column of table 1, I present the results of estimating the model with the addition of six time dummies. The inclusion of these time dummies alters the estimates and improves substantially the value of the overidentification test. The more significant change is the reduction in the habits coefficient ϕ . This indicates that one should control for the presence of macroeconomic effects as well as for unobserved heterogeneity, to correctly measure the degree of state dependence. Indeed, part of the effect of the lagged-hours variable may be due to business-cycle effects. However, the estimated effect of the previous year-hours decision

remains important and significant, having allowed for year dummies as well as for individual dummies. If the link of unemployment to hours does not work through the wage but is correlated with the incidence of labor-market constraints, then the state unemployment would be a suspect instrument. Ham (1986) found evidence in favor of this hypothesis using a utility function that was separable over time and individual's unemployment spells in the hours equation. In the context of my model, I included the state unemployment as an additional variable in the hours equation with and without time dummies to see whether there was a link to hours not captured by wages or lagged hours. In the model without time dummies, the estimated coefficient on the state unemployment variable was -0.03 with a standard error of 0.40 , whereas in the model including time dummies I obtained 2.94 with a standard error of 6.25 . In the first case, the other coefficients remained largely unaffected, while, in the second, point estimates were in general on line with those reported on table 1 but standard errors were bigger, rendering all coefficients insignificant with the exception of γ_b (the estimate of ϕ in this case was the next most significant coefficient with a value of 0.13 and a standard error of 0.10). Also notice that, given the instrumentation of h_{it-1} , the positive and significant estimates of the habits coefficient ϕ shown in table 1 support a nonseparable model as opposed to a separable model with serially correlated errors. The overidentification test is also satisfactory.

Table 2
Summary Measures of the Estimated Distributions
of the Elasticities

	Values
ϵ_{it} :	
Mean of the estimated sample elasticities	.0503
Standard deviation of the mean	.0098
Evaluation at sample means	.0822
Upper quartile	.1076
Median	.0682
Lower quartile	.0307
Skewness	-63.6315
η_{it}^a :	
Mean of the estimated sample elasticities	-.1470
Standard deviation of the mean	.0158
Evaluation at sample means	-.1093
Upper quartile	-.0755
Median	-.1084
Lower quartile	-.1476
Skewness	-59.5643
η_{it}^b :	
Mean of the estimated sample elasticities	.0273
Standard deviation of the mean	.0218
Evaluation at sample means	-.0184
Upper quartile	.0304
Median	-.0144
Lower quartile	-.0605
Skewness	67.273

Since the elasticities are data dependent, a more detailed account of their sample variability, for the base model in the first column of table 1, is given in table 2. This table shows that the empirical distributions of the estimated elasticities are nonnormal and, in particular, asymmetric, the median being therefore a more reliable measure of the position for the distribution of the estimated elasticities than the arithmetic mean. The estimated effects of permanent wage changes are negative, as one might expect, a difference in the level of the wage profile having a bigger negative impact on hours than a difference in the growth of wages.

One should conclude that, although these results strongly reject the hypothesis of intertemporal separability, they suggest that estimating more general nonseparable models does not alter the widespread finding of a small intertemporal substitution elasticity in life-cycle models of labor supply estimated from individual male data as, for example, in the studies of MaCurdy (1981) and Altonji (1986), which also use the Michigan panel.

V. Conclusion

In this article, I have estimated a nonseparable intertemporal model of labor supply to see whether the usually maintained intertemporal separability is sustained by the data and how relaxing separability affects the estimates of the elasticities. I specified a rational-habits model that produced sensible results.

My results, using a sample of men from the Michigan panel, strongly support the importance of past hours in determining current hours decisions. This is true even allowing for individual as well as time-specific effects. These results call for more research on intertemporally nonseparable models and indicate that the usual intertemporal models that assume separability over time are misspecified. Allowing for nonseparable preferences, however, does not seem to alter significantly the small size of the intertemporal substitution elasticity for men found in the literature that assumes intertemporal separability.

Appendix

Table A1

Statistics of Relevant Variables ($N = 785$ Men)

Variable	Mean	SD
Real earnings	10,483.01	6,100.12
Annual hours worked _{<i>t</i>}	2,328.16	579.99
Annual hours worked _{<i>t-1</i>}	2,337.87	573.49
Real hourly wages	4.66	2.72
Age	41.06	8.49
Number of children	1.79	1.47
A_{it}	1,864.33	3,166.77
$\hat{\alpha}_{it}$	4.32	2.75
δ_{it}	.09	.66
$\hat{\alpha}_{it} + \delta_{it}$	4.67	2.67

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